CHRONOS RESEARCH PROJECT | WORKING PAPER 01

Sticky Probabilities

A Mathematical Study of Polymarket Price Dynamics

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Epistemic status. Paper-ready working draft based on chronological development, a no-refit later-cohort replication, an official five-minute measurement falsifier, and one retained protected negative. No protected finding and no claim of executable alpha.

Abstract

We study registered Polymarket YES-probability samples as bounded, event-linked time series. The main development corpus contains 803,862 hourly observations from 562 markets in 157 events. We also use a separately registered later cohort and an independently acquired official five-minute Central Limit Order Book panel. A smooth constant-intensity diffusion is a poor description of the registered process: interior hourly increments are exactly zero 62.40% of the time, raw kurtosis is 157.10, adjacent nonzero signs reverse 62.21% of the time, and 24-hour variance scales far below the diffusion benchmark. Update intensity depends strongly on elapsed quiet time.

We compare bounded Gaussian diffusion with memoryless, state-conditioned, and state-duration/path zero-inflated marked-jump laws in chronological whole-event evaluation. The full marked-jump law improves development event-macro negative log score by 7.74% over the state-only hurdle and wins 112 of 113 event comparisons. Frozen without refitting, it improves 7.88% on a later cohort and wins all 27 eligible events against both state-only and diffusion comparators. A continuous logistic arrival hazard separately improves later-cohort event Brier by 33.38% and wins 27 of 27 events. Its formally frozen implementation nevertheless fails protected promotion because only 65 of 100 declared decisions execute. We conclude that the registered process is best described as a bounded, sticky, zero-inflated marked-jump process with state-, duration-, and path-dependent components. This is a result about registered observations, not a latent belief SDE, executable price process, or profitable strategy.

Executive findings

Process family	Bounded zero-inflated marked jumps dominate a simple bounded Gaussian diffusion.
Strongest component	Quiet duration and recent inactivity materially predict whether the next registered sample changes.
Robustness	The structure recurs in later events, at five-minute frequency, away from boundaries, in log-odds, and across maturity and contract-structure strata.
Trading implication	None yet. Update timing is not direction; sampled probabilities are not executable quotes; spread, fees, latency, and depth remain separate hurdles.

Scope. "Polymarket" identifies the venue studied. This research is independent and is not affiliated with or endorsed by Polymarket.

1. Research question and empirical design

A binary YES share pays $X \in \{0, 1\}$ at resolution. Under sufficient no-arbitrage and market- completeness assumptions, a price can be written as a discounted conditional expectation under a pricing measure. Those assumptions do not make a recorded Polymarket field the objective event probability, average trader belief, executable midpoint, or physical-measure expectation. The paper therefore asks a narrower and testable question:

> Which stochastic-process family best describes and predicts the *registered sampled YES-probability process*, and which restrictions from conventional financial time-series theory survive after respecting bounded payoffs, event dependence, finite maturity, and temporal legality?

1.1 Retained evidence surfaces

Evidence surface	Events	Markets	Observations	Resolution	Role
Fresh development cohort	157	562	803,862	Hourly	Description and chronological model development
Later successor cohort	29	147	42,549[a]	Hourly	Unchanged no-refit replication
Official CLOB panel	10	10	19,258	Five minute	Source and sampling-clock falsifier
Formal arrival schedule	111	–	231 decisions	Sparse	Governed development evaluation
Original final block	47	221	Unavailable	Hourly	Protected and unexposed in the opening design

[a] Count for the later process-signature replay. Corpus identities and exact receipts appear in Appendix A.

The production snapshot was acquired through the read-only `human_ro` database role and retained by content identity. An experiment may use only registered fields exposed through its capability. The data foundation records the exact source digest, cursor vintage, selection, partition, feature accesses, source code, resource envelope, and output receipt.

1.2 Time legality and dependence

At a decision time t, a feature is legal only if its `available_at` $\leq t$. Labels use the first later registered sample after the declared horizon. Parameters are fitted using earlier event groups and evaluated on later whole events. Training gives each represented event equal total weight; reporting includes observation-weighted and event-macro scores. This prevents a single event with many related contracts or a long path from silently defining the result.

The design distinguishes three clocks:

- observation ordinal;
- elapsed UTC time between retained observations; and
- remaining time to recorded market close.

Headline one-step diagnostics require elapsed gaps between 0.75 and 1.25 hours. Multi-step variance diagnostics require the elapsed clock to fall within 25% of the stated horizon.

1.3 What is deliberately not assumed

Registered probability samples are not silently reclassified as bids, asks, trades, midpoints, or latent beliefs. Related markets are not IID rows. Realized payout may diagnose terminal composition but cannot enter a predictor. A development success cannot access protected values, and a failed protected system cannot be repaired on the same exposed holdout.

Interpretation boundary. The estimand is the registered observation process. Claims about execution, profit, objective probability, or a latent continuous-time efficient price require additional data and tests.

2. Candidate stochastic-process laws

Let $P_t \in [0,1]$ be the registered YES probability and $\Delta P_{t+1} = P_{t+1} - P_t$ a regular one-hour increment. Let $\mathcal{F}_t$ contain the legal state: probability band, time to close, quiet duration, recent zero fraction, previous nonzero sign, and previous nonzero magnitude.

2.1 Bounded diffusion comparator

The baseline is a zero-drift Gaussian interval law with state-dependent scale:

$$\Delta P_{t+1} \mid \mathcal{F}_t \sim \mathcal{N}\big(0,\ \sigma^2(c_t)P_t(1-P_t)\big)\,, \tag{1}$$

where c_t is a fixed probability/time-to-close cell. The bounded-state factor allows local variance to shrink near zero and one. It is intentionally a simple test of smooth, conditionally Gaussian accumulation rather than a claim that every continuous latent-price model has this form.

2.2 Zero-inflated marked-jump family

The hurdle family separates whether a registered change occurs, its sign, and its magnitude:

$$\Pr(\Delta P_{t+1} = 0 \mid \mathcal{F}_t) = 1 - q(\mathcal{F}_t), \tag{2}$$

$$\Pr(\Delta P_{t+1} > 0 \mid \Delta P_{t+1} \neq 0, \mathcal{F}_t) = s(\mathcal{F}_t), \tag{3}$$

$$\log|\Delta P_{t+1}| \mid \Delta P_{t+1} \neq 0, \mathcal{F}_t \sim \mathcal{N}\big(\mu(\mathcal{F}_t), \tau^2(\mathcal{F}_t)\big)\,. \tag{4}$$

The full law uses probability and quiet-duration cells for q, probability and previous nonzero sign for s, and probability plus previous nonzero magnitude for (μ,τ). Every cell shrinks toward the event-equal global law with one event of prior weight. Nested alternatives remove path variables (state-only), remove state and duration (memoryless), or remove the point mass entirely (diffusion).

For a nonzero movement x, every model is scored on the same 0.001-wide observed-price interval:

$$\Pr(x \mid \mathcal{F}_t) = q(\mathcal{F}_t)\Pr(\operatorname{sign} x \mid \mathcal{F}_t)\left[\Phi\left(\frac{\log(|x|+\epsilon)-\mu}{\tau}\right) - \Phi\left(\frac{\log(\max(10^{-6},|x|-\epsilon))-\mu}{\tau}\right)\right], \tag{5}$$

with $\epsilon = 0.0005$. A zero receives point mass $1 - q(\mathcal{F}_t)$. The primary score is negative log interval probability, averaged both by observation and by event.

2.3 Continuous update-arrival hazard

The strongest component model predicts $Y_{t+1} = 1[\Delta P_{t+1} \neq 0]$ using regularized logistic regression:

$$\operatorname{logit}\Pr(Y_{t+1}=1 \mid \mathcal{F}_t) = \beta_0 + \beta_p f(P_t) + \beta_T f(T_t) + \beta_q \log(1+Q_t) + \beta_z Z_{t,24}, \tag{6}$$

where Q_t is the current quiet run and $Z_{t,24}$ is the fraction of unchanged regular samples in the prior 24 hours. Its Brier score is

$$\mathrm{BS} = n^{-1}\sum_{i=1}^{n}(\widehat{q}_i - Y_i)^2. \tag{7}$$

2.4 Falsifiable restrictions

A smooth diffusion is contradicted if regular interior increments exhibit at least 25% exact zeros, kurtosis at least 20, or a 24-hour variance ratio outside $[0.75, 1.25]$. A memoryless clock is questioned when next-move rates differ by more than a factor of two between freshly active and 24-hour-quiet states. Directional predictability requires separate out-of-time gates and cannot be inferred from volatility or arrival prediction.

3. Descriptive evidence: sticky, discontinuous, and path dependent

Signature	Earlier cohort	Later cohort	Interpretation
Interior regular-hour zero mass	62.40%	44.85%	Strong but regime-dependent stickiness
Raw regular-increment kurtosis	157.10	55.01	One Gaussian increment law is inadequate
24-hour variance ratio	< 0.75 in 142/151 events	0.230 pooled; < 0.75 in 24/26	Variance does not scale linearly in clock time
Nonzero sign reversal	62.21%; $> 50\%$ in 134/152	$> 50\%$ in 21/26	Registered marks are negatively dependent
Long-quiet/active move-rate ratio	0.025; < 0.5 in 83/83	0.074 event mean; < 0.5 in 7/7	Arrival depends on elapsed quiet time
Boundary persistence, low/high	98.99% / 97.07%	98.16% / 99.12%	Terminal boundaries materially shape paths

Table 1: Fixed process signatures. Exact levels vary; the structural directions recur.

Among all consecutive probability samples, 69.62% are unchanged. For regular interior hourly increments, the median absolute change is zero; the 90th, 99th, and 99.9th percentiles are 1, 9, and 32 cents. Variance ratios decline from 0.723 at two hours to 0.536, 0.393, and 0.235 at 4, 8, and 24 hours, rather than remaining near one as independent diffusion increments would imply.

3.1 The update clock remembers inactivity

Prior unchanged samples	Rows	Next-hour move rate	Mean absolute change
0	236,973	68.24%	1.173 cents
1–5	199,150	28.90%	0.340 cents
6–23	145,703	9.50%	0.116 cents
24+	199,562	1.72%	0.025 cents

Table 2: Observed next-sample activity conditional on the preceding quiet run.

The 24+-quiet move rate is only 2.53% of the freshly active rate. This rejects a constant- intensity update clock for registered observations. It does not by itself distinguish economic inactivity from a sampling, rounding, or microstructure mechanism.

3.2 Boundaries and sampling frequency do not explain the result away

Strict-interior probability and log-odds representations retain high zero mass, raw kurtosis above 100, long-quiet hazard collapse, and sub-diffusive aggregation. Separate early and late maturity regimes agree in direction. Aggregating to six hours weakens but does not remove stickiness or sub-diffusion.

On 19,012 regular official five-minute increments, interior zero mass is 78.00%, kurtosis is 203.64, nonzero reversal is 67.87%, and the 24-hour variance ratio is 0.067. Next-sample movement falls from 55.95% with no prior quiet sample to 0.20% after 24 quiet hours. The official path matches 1,580 eligible hourly samples within ten minutes; 99.49% match exactly and mean absolute difference is 0.0000487. Coarse hourly sampling alone therefore cannot account for the main signatures on this preselected panel.

The evidence rejects the joint descriptive restrictions of a smooth Gaussian diffusion and a memoryless update clock. The most stable empirical facts are point mass at zero, duration-dependent arrival, bounded-state persistence, sign reversal, and sub-diffusive aggregation.

4. Chronological predictive comparisons

All comparisons fit earlier whole events and score fixed later-event blocks. The complete-law experiment makes 566,501 out-of-time predictions over 113 events.

4.1 Complete one-hour observation law

Model	Development observation NLL	Development event NLL	Later event NLL
Bounded state-dependent Gaussian diffusion	3.511	3.578	4.582
Memoryless zero-inflated marked jump	1.865	2.214	3.240
State-conditioned zero-inflated marked jump	1.782	2.104	3.116
State + duration/path marked jump	**1.585**	**1.941**	**2.870**

Table 3: Lower negative log interval score is better. The later law is replayed without refitting.

The full law improves development event-macro NLL by 7.74% over state-only, wins 112 of 113 event comparisons against state-only, and wins 113 of 113 against diffusion. Its advantage appears in both chronological halves. The exact fitted law is then frozen and replayed without fitting on a later cohort: it improves 7.88% over state-only and wins all 27 eligible events against both state-only and diffusion. Absolute scores deteriorate across cohorts, indicating a stable structural family with regime-dependent parameters rather than one universal calibration.

4.2 Which components contribute?

Target and added information	Development gain	Later replay	Interpretation
Move arrival: continuous quiet duration and recent zero fraction	27.63% event-Brier	33.38%; 27/27 wins	Strongest component
Nonzero sign: prior-path information	3.96% event-Brier	Not separately replayed	Modest reversal information
Nonzero log magnitude: prior magnitude path	20.61% event-MSE	Not separately replayed	Strong magnitude clustering
Heavy-tailed residual after hurdle decomposition	Rejected	Not replayed	Raw tails need not imply heavy residuals

Table 4: Component evidence relative to fixed state-only comparators.

The frozen continuous arrival hazard in Equation 6 is reconstructed from retained development coefficients and replayed on 17,671 later predictions:

Later-cohort hazard	Observation Brier	Event Brier	Event log loss
State-only logistic	0.24319	0.26106	0.71628
State + continuous inactivity	**0.16477**	**0.17392**	**0.52634**

The full hazard wins all 27 eligible events and both chronological halves. Its mean predicted move rate is 0.4129 versus an observed 0.4534, an absolute calibration gap of 0.0405. This is strong no-refit evidence that inactivity is predictive of the registered update process.

Economic boundary. Forecasting whether a sampled price changes is not the same as forecasting its direction or earning a return. A useful arrival model may eventually improve trade eligibility, execution scheduling, or simulation without being alpha by itself.

5. Formalization and the protected negative

Chronos separates exploratory or chronological development evidence from a frozen scientific claim. A candidate must pass fixed development gates before an independent freeze seals its source, features, fitting recipe, decision schedule, metrics, coverage requirements, and protected plan.

5.1 Coarse compression is rejected

The first formal candidate compresses the complete law into a sparse binary arrival forecast using fixed probability and quiet-duration cells. It narrowly misses its absolute gates:

Formal check	Result	Required	Decision
Observation Brier	0.202002	≤ 0.200000	Fail
Event-macro Brier	0.207234	≤ 0.200000	Fail
Early event Brier	0.194038	≤ 0.220000	Pass
Late event Brier	0.220194	≤ 0.220000	Fail
Calibration gap	0.048827	≤ 0.050000	Pass

The result is retained as a rejection rather than rounded into success. A descended candidate keeps the target, temporal machinery, event-aware weighting, and sparse schedule but substitutes the already replicated continuous inactivity features. It passes governed development with 202 out-of-fold predictions across 100 events: observation Brier 0.18256, event Brier 0.19018, early and late event Brier 0.18741/0.19296, and calibration gap 0.00463.

5.2 One protected resolution

The descended system is frozen independently and resolved exactly once:

Protected check	Result	Required	Decision
Brier on completed rows	0.189254	≤ 0.200000	Pass
MAE on completed rows	0.394489	≤ 0.400000	Pass
Coverage	0.650000	≥ 0.950000	Fail
Failure rate	0.350000	≤ 0.050000	Fail
Maximum error concentration	0.054852	≤ 0.050000	Fail

Table 5: The exact frozen system is rejected despite promising completed-row accuracy.

Thirty-four decisions lack the fixed 100-row legal training window and one lacks a legally available outcome. These are system failures, not removable missing data. Because coverage may be non-random, reporting only the 65 completed predictions would overstate generalization. The validator therefore rejects the candidate and retains the protected negative. That holdout is now exposed and cannot be reused to develop a replacement.

> The protected failure is part of the result. It distinguishes a scientifically useful process component from an operationally complete predictive system. Accuracy conditional on execution does not compensate for failed decisions.

5.3 What the result does and does not establish

The complete density law has strong chronological and no-refit later-cohort evidence. The continuous hazard has strong later replay and formal-development evidence. Neither is a protected finding. The exact protected arrival system is terminally rejected. No test on these sampled probabilities evaluates spread, fees, latency, slippage, market impact, or directional P&L.

6. Financial interpretation and conclusion

Financial idea	Assessment	Prediction-market modification
Proper scoring and chronological validation	Transfers directly	Cluster and weight by event; retain the complete trial history
Local/state-dependent volatility	Transfers	Respect bounded state, semantic resolution, and terminal persistence
Duration and point-process models	Transfers strongly	Intensity depends on inactivity, state, semantics, and observation mechanism
Volatility/mark clustering	Transfers	Separate arrival clustering from sign and conditional magnitude
Heavy-tailed returns	Transfers conditionally	Decompose point mass and conditional scale before choosing a residual law
Martingale benchmark	Useful but conditional	Pricing measure, physical probability, belief, sample, and executable price differ
Constant Brownian diffusion	Fails descriptively	Requires point mass, bounded state, jumps, and irregular information time
Independent signs	Fails for registered marks	Reversal may reflect bounce, rounding, aggregation, or belief correction
Black–Scholes replication	Does not transfer directly	No continuously traded underlying and no standard completeness argument

The best-supported law for registered Polymarket samples is a bounded, zero-inflated marked-jump process with state-, duration-, and path-dependent intensity and marks. "Jump" means a nonzero registered change; it is not proof that the latent efficient probability is discontinuous. The observation may combine information arrival, trading and quote sampling, rounding, stale values, bid/ask mechanics, and participation.

Representation tests make the conclusion narrower but stronger. Boundary absorption matters but does not create the entire effect. Strict-interior log-odds preserves the qualitative law. Six-hour aggregation moves the series toward diffusion-like scaling but does not eliminate zero inflation or sub-diffusion. Event breadth, contract lifetime, and retained path depth alter parameter levels without reversing the structural signatures. The defensible object is therefore a scale- and regime-dependent sampled process, not one coordinate-invariant latent SDE with universal parameters.

6.1 Research implications

- Simulators and risk models should represent an update clock separately from direction and magnitude.
- Time since the last change is a first-class state variable, not merely a missing-data counter.
- Evaluation should split by whole events and chronology; row-level random cross-validation is inappropriate.
- A directional model should use the arrival hazard as an eligibility component, then prove incremental sign or return information separately.
- Tradability requires side-specific executable prices, spread, fees, latency, liquidity, and abstention. Sampled-price accuracy is insufficient.

6.2 Conclusion

Conventional finance remains useful when rewritten around a binary, finite-maturity, event-linked payoff. Proper scoring, state-dependent volatility, duration models, and marked point processes transfer. Constant diffusion, independent signs, and direct Black–Scholes-style replication are poor defaults. Across broad chronological development and an unchanged later replay, the leading sampled-process family is a state-, duration-, and path-conditioned zero-inflated marked jump.

The paper's strongest contribution is structural: Polymarket observations are sticky because the arrival of registered changes is itself stateful, and subsequent nonzero signs and sizes retain path information. The exact protected arrival system nevertheless fails its predeclared coverage and concentration standards. The correct conclusion is neither "markets are random" nor "we found alpha." It is that a useful stochastic-process foundation has been identified, replicated, bounded by a protected failure, and made reproducible enough to support the next generation of execution-aware and alternative-data research.

Bottom line. Chronos found a reproducible mathematical description of how registered Polymarket probabilities move. It has not yet found a protected or executable trading edge.

Appendix A Data authority, partitions, and reproducibility

Corpus identities

The opening 1,000-contract corpus contains 235 events and 1,180,437 sampled probabilities. The successor acquisition contains 1,509 contracts, 321 events, and 1,585,740 observations. Its content identity is:

```
polymarket-corpus:184b83f7f8012da4ca9116c8b6bad2076570ad5f0bada95b22615ee2f5cceae8
```

The successor partition contains 29 wholly fresh development events (147 markets), 227 embargoed ancestor events (1,005 markets), and 65 protected events (357 markets). Expansion adds later data without changing an earlier experiment's bytes, selection, identity, or partitions.

Scientific transition record

For each public run, Chronos retains:

- research question and exact authored source digest;
- dataset and partition identities;
- actual field accesses and temporal-availability checks;
- resource envelope, worker outcome, and deterministic receipt;
- development, freeze, protected, reporting, and negative-result transitions;
- failed trials and superseded artifacts rather than only favorable outputs.

Core evidence is indexed in `paper/evidence-index.md`. Deterministic figure construction and replay commands are recorded in `paper/reproducibility.md`. The protected source is not to be rerun: its holdout was exposed once and the exact frozen system was rejected.

Appendix B Robustness and heterogeneity details

Fixed view	Increments	Zero rate	Kurtosis	Event-equal 24h ratio
All probability rows	781,388	69.73%	157.10	0.358
Strict interior probability	519,495	62.76%	103.19	0.376
Strict interior log-odds	519,495	62.76%	104.99	0.403
More than 168 hours to close	701,428	69.40%	151.06	0.338
Zero to 168 hours to close	79,960	72.54%	138.44	0.512

Table 6: Fixed representation and maturity falsifiers.

Whole-event stratum	Zero mass	Variance ratio	Reversal	Quiet ratio
One-market events	0.577	0.403	0.588	0.084
At least five markets	0.527	0.267	0.598	0.032
Lifetime $\leq$ 30 days	0.477	0.293	0.581	0.038
Lifetime 31–180 days	0.605	0.425	0.594	0.078
Lifetime $>$ 180 days	0.623	0.228	0.643	0.035
Fewer than 500 points	0.458	0.315	0.556	0.014
500–1,999 points	0.586	0.394	0.601	0.070
At least 2,000 points	0.631	0.248	0.635	0.053

Table 7: Event-equal structural heterogeneity. Every headline-eligible stratum remains on the predeclared side of all four thresholds.

Exact parameter levels vary materially. The evidence supports recurrence of a process family, not parameter invariance across political, economic, sports, or other semantic regimes.

Appendix C Calibration and alternative-data null results

The research campaign also tested whether descriptive or predictive structure immediately yields directional edge. These studies use separate data authority and do not strengthen the process-law claim.

Calibration lead

At 24, 72, 168, and 336 hours before close, sampled mean probabilities exceeded eventual YES rates by approximately 4.1–9.2 percentage points under market weighting and by 4.1–7.4 percentage points under event-macro weighting. The sign replicated in a later cohort at 72, 168, and 336 hours, although long-horizon coverage was thin. This resembles a favorite–longshot calibration pattern but does not establish mispricing: semantic mix, related outcomes, selection, and sampled versus executable price remain live alternatives.

Signed flow and fill semantics

An apparent raw 15-minute flow association of roughly −0.46 collapsed to approximately +0.04–+0.05 after transaction direction was handled correctly. A separate audit found that Polymarket's retained fill table is a per-party ledger: maker and taker legs must not be counted as independent market trades. Canonical external studies now retain only the transaction-side record and convert YES/NO fills into a consistent YES-equivalent price and taker direction. The invalid interpretation is preserved as a negative data lesson.

Brazil polling experiment

Thirty-one exact national Lula/Flávio runoff poll releases were reconstructed from immutable publication history, producing 25 decision events and 17,570 canonical Polymarket fills. Features included latest poll change, rolling-three median change, surprise against the previous five releases, and same-pollster change. Follow and fade rules were tested at 0, 2, and 4 percentage- point thresholds over 24, 72, and 168 hours.

Every rule with at least five executable pairs was negative after conservative fees. The broad 72-hour fade earned $10.69 gross but lost $18.97 net over seven trades. The least-negative thresholded rule lost $14.70 over five trades. The hypothesis was rejected before protected formalization.

Current alternative-data direction

The next independent question uses another prediction venue only as a time-legal external predictor: do earlier Kalshi price, activity, disagreement, or shock features improve later Polymarket direction, magnitude, or move-arrival forecasts? This is not treated as an arbitrage study. Any candidate must be evaluated chronologically and then translated to side-specific Polymarket execution with spread, fees, liquidity, latency, abstention, and directional P&L.

Appendix D Claim ledger

Claim	Status	Evidence boundary
Smooth constant-intensity diffusion describes registered hourly samples	Rejected	Zero mass, tails, time scaling, duration dependence
State-duration/path marked-jump family improves predictive density	Strong development + later replay	Not protected; targets registered samples
Continuous inactivity improves next-update probability	Strong development + later replay	Exact protected implementation rejected
Boundaries alone explain stickiness	Rejected as sole explanation	Strict-interior and log-odds robustness
Hourly sampling alone creates the process signatures	Rejected on selected panel	Official five-minute source
Observed sign reversal is executable mean reversion	Not established	Quotes, latency, spread, and depth absent
Sampled probabilities are universally miscalibrated	Not established	Development lead with semantic and selection rivals
Chronos has found tradable alpha	No	All cost-aware directional ideas currently rejected

Appendix E Limitations

- Registered samples are not identified as trades, quotes, midpoints, latent beliefs, or executable prices.
- The five-minute panel contains ten preselected markets and is a measurement falsifier, not a venue-wide quote study.
- Related contracts and semantic groups create dependence that event grouping mitigates but cannot eliminate.
- The diffusion comparator is deliberately simple; rejecting Equation 1 does not reject every latent continuous-time model with endogenous sampling and microstructure noise.
- Representation and contract-structure robustness are fixed development falsifiers, not independent protected replications.
- The complete density law cannot currently enter the scalar protected evaluator as a governed distribution-valued log score.
- One protected candidate was evaluated and rejected. Its 35% failed decisions prevent inference from completed-row accuracy to the full declared schedule.
- No result establishes causal information arrival, objective-probability equality, Sharpe ratio, or net profit after all execution costs.

Appendix F References

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Reproducibility statement

The manuscript is a presentation layer over retained Chronos receipts. Numerical claims map to content-addressed evidence in `paper/evidence-index.md`; replay commands and non-rerunnable protected boundaries are documented in `paper/reproducibility.md`. Where a result is development-only, later-replayed, formally rejected, or protected-negative, the status is stated explicitly rather than collapsed into a single significance label.

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